

MD SHIFATUL AHSAN APURBA

+1 (205) 365-0219 · BIRMINGHAM, AL

✉ MAPURBA@UAB.EDU ·  MAPURBA ·  APURBA-ONLINE ·  SCHOLAR

OBJECTIVE

AI and data science researcher and Ph.D. candidate in Biomedical Health Informatics and Data Science at the University of Alabama at Birmingham, with research interests in generative modeling, computer vision, representation learning, and large language models. Research focuses on developing AI-driven methods for image reconstruction, multimodal learning, and evidence-based analysis. Seeking research and applied AI/ML opportunities to contribute to reliable, efficient, and impactful intelligent systems across biomedical, scientific, and broader real-world applications.

EDUCATION

Ph.D. Candidate, Biomedical Health Informatics Expected April 2028

Department of Biomedical Informatics and Data Science

University of Alabama at Birmingham

CGPA: 4.00 out of 4.00 (Current)

Master of Computer Science and Engineering, BRAC University August 2024

CGPA: 3.73 out of 4.00

Bachelor of Computer Science and Engineering, BRAC University August 2022

CGPA: 3.74 out of 4.00

SKILLS

Technical Skills Python, Java, ASP.NET, SQL, Machine Learning, Power BI, Power Automate, Power Apps

Advanced AI and Data Analysis TensorFlow, PyTorch, Scikit-learn, OpenCV, Neural Networks, Transformers, Generative Models (Diffusion Models, GANs, MeanFlow Models), Reinforcement Learning, Large Language Model Fine-Tuning, Prompt Tuning and Instruction Tuning of LLMs, Vision-Language Models (OpenAI CLIP, SigLIP), Medical Image Processing, Radiomics, Pandas, NumPy, Matplotlib, Data Pre-processing, Model Training, Evaluation, and Optimization

Soft Skills Leadership, Communication, Teamwork, Problem-Solving, Adaptability, Mentorship

Others Adobe Photoshop, Adobe Illustrator, Adobe Premier Pro

EXPERIENCE

Graduate Research Assistant (*Part-time*) Aug 2024 - Present

Dept. of Biomedical Informatics and Data Science, University of Alabama at Birmingham

Birmingham, AL

- Conduct research on generative AI methods for CT image standardization, reconstruction, and medical image analysis.
- Build and evaluate deep learning pipelines using clinical imaging datasets, focusing on image quality, structural similarity, and radiomic feature consistency.
- Support literature review, experimental design, data analysis, and manuscript preparation for biomedical imaging research projects.

Specialist, IT Core Applications (*Full-time*) Feb 2024 - Jul 2024

Robi Axiata PLC

Dhaka, Bangladesh

- Upgraded capacity by 20% for CBS voucher card project implementing advanced SUSE Linux system integrating with APIs and Gauss DB in the system.
- Revamped CDR (call details record) mediation system while saving 30% storage.

- Automated monthly capacity planning and reporting for 3 major charging systems.

Debate Coach (*Part-time*)
Maple Leaf International School

Oct 2022 - Jul 2024
Dhaka, Bangladesh

- Teaching debate to 150+ students (Grade 6-12) involves introducing structure, research skills, persuasive and logical reasoning, public speaking, effective rebuttals, time management, critical listening, respectful dialogue, practice debates, debate etiquette, ethics, and evaluation.

Associate Software Engineer (*Full-time*)
British American Tobacco Bangladesh

Jan 2023 - Jan 2024
Dhaka, Bangladesh

- Launched 21 projects using Power Platform, ASP.NET, PHP Frameworks and ReactJS.
- Saved 11,000+ man-hours and significant operating costs per year including environmental sustainability with automated and paperless initiatives.
- Initiated citizen development program and impacted 700+ employees.

Intern, Digital Business Solutions (*Internship*)
British American Tobacco Bangladesh

Sep 2022 - Dec 2022
Dhaka, Bangladesh

- Coordinated IT projects collaborating with 3 different end-market.
- Arranged 10+ IT workshops for internal employees.
- Built company's guesthouse reservation and management app.

Undergraduate Teaching Assistant (*Part-time*)
School of Data and Science, BRAC University

Nov 2021 - Aug 2022
Dhaka, Bangladesh

- Taught Matlab, Mathematica, and Python and graded scripts of 250+ Students.

Intern, Core Network Service Management (*Internship*)
Banglalink Digital Communication Ltd.

Oct 2021 - Dec 2021
Dhaka, Bangladesh

- Planned data-driven network routing for three nodes spanning two divisions within the country.

RESEARCH AND PUBLICATION

- [1] Veysel Karani Baris, Alexandra Frost, Michael Sainlaire, Jin Chen, Minakshi Shukla, **Md Shifatul Ahsan Apurba**, Ann Hurley, and Patricia C Dykes. Exploring primary care providers' perspectives on clinical decision support for venous thromboembolism diagnosis: a qualitative study. *BMJ Open*, 16(9):e122559, 2026. [Read]
- [2] **Md Shifatul Ahsan Apurba**, Md Selim, and Jin Chen. RA-CMF: Region-Adaptive Conditional MeanFlow for CT Image Reconstruction. Presented at *International Conference on Intelligent Biology and Medicine (ICIBM)*, 2026. [Read]
- [3] Mysha Samiha Priota, SK Fahema Mahajabin, Maharun Fahamida Maysha, Fazley Rabbi, Md. Imran Hossen, **Md Shifatul Ahsan Apurba**, Md. Arman Hosen Patoary, Md. Yasin Arafat Tamim, and Tauhidur Rahman Sakib. A Hybrid DenseNet-Vision Transformer Framework for Diabetic Retinopathy Classification. *Proceedings of the International Conference on Cybersecurity, Digital Forensics, and AI Applications (ICCSDFAI)*, pp. 1255–1260, IEEE, 2026. [Read]
- [4] Alexandra C. Frost, Veysel K. Baris, Michael Sainlaire, Jin Chen, Minakshi V. Shukla, **Md Shifatul Ahsan Apurba**, Ann C. Hurley, and Patricia C. Dykes. A model to guide successful living with venous thromboembolism (VTE): Recommendations derived from the qualitative phase of a study of diagnostic Delay of VTE (DOVE). *Thrombosis Research*, 258:109585, 2026. [Read]
- [5] Sanjib Kumar Sen[†], **Md Shifatul Ahsan Apurba**[†], Anika Priodorshinee Mrittika, Md. Tawhid Anwar, A.B.M. Alim Al Islam, and Jannatun Noor. Unveiling shadows: A data-driven insight on depression among Bangladeshi university students. *Heliyon*, 11(1):e41110, 2025. [Read]

- [6] **Md Shifatul Ahsan Apurba**, P. Saha, M. S. Hossain, K. A. R. Pranta, M. M. Alam, and A. A. Rasel. Development of an OCR-Based System for Reading Text in Dull Conditions. Presented at the *IEEE R10 Humanitarian Technology Conference (HTC)*, 2023.
- [7] **Md Shifatul Ahsan Apurba**. A Comprehensive Survey on Optimizing Storage Models, Data Layouts, and System Catalogs. *International Journal of Distributed Computing and Technology*, 9(2):1–8, 2023. [Read]

† Co-first authors.

PROJECT

Personal Project. An Interactive Research Workspace for Provenance-Preserving Biomedical Evidence (Visit here) Developed an LLM-powered interactive research workspace for provenance-preserving biomedical evidence exploration. The system integrates multiple biomedical APIs, including RxNorm, DailyMed, openFDA, ClinicalTrials.gov, ChEMBL, Open Targets, and PubChem, with uploaded scientific literature to support natural-language evidence retrieval, cross-source reasoning, and structured synthesis. It separates LLM-based interpretation from deterministic evidence composition to improve reproducibility and maintain traceable source provenance.

EXTRA-CURRICULAR ACTIVITIES

Debating. Actively participated in and judged national and international debate tournaments, breaking at major events such as ABP 2021 and ADTI Indo Pacific Pre-WUDC 2021, achieving 2x quarter-finalist positions at Uhuru Worlds, and securing over 15 championship wins.

Filmmaking. Involved in filmmaking with achievements including the nomination of Bakshobondi as the 2nd Best Film at International Children’s Film Festival 2017 and as the Best Edited Film at Sylhet International Film Fest 2017, along with the official selection of Oddhayer Sesh at International Children’s Film Festival 2015.

Hackathon. Participated in several hackathons, including Robi Datathon 2.0, where placed in the top 20%, and developed a Covid Update and Alerts application during the pandemic, earning the 2nd runners-up award at MIST National ICT Fest 2021.

LEADERSHIP

- Led the BRAC University Debate Club as President, facilitating over 500 students and providing necessary resources, which helped the club secure the World University Debating Championship 2022 title. Convened 1 international and 3 national tournaments during my tenure.
- Led student management for residential semester at BRAC university, initiating an automated meal management system that streamlined operations. Coordinated cultural events such as RS Day and Cultural Night to enhance student engagement.
- Served as a mentor for the first-year advising team at BRAC University, providing guidance to over 50 students on course selection, academic planning, and advising support. Fostered a supportive environment to help students navigate their academic journey and make informed decisions for their future.